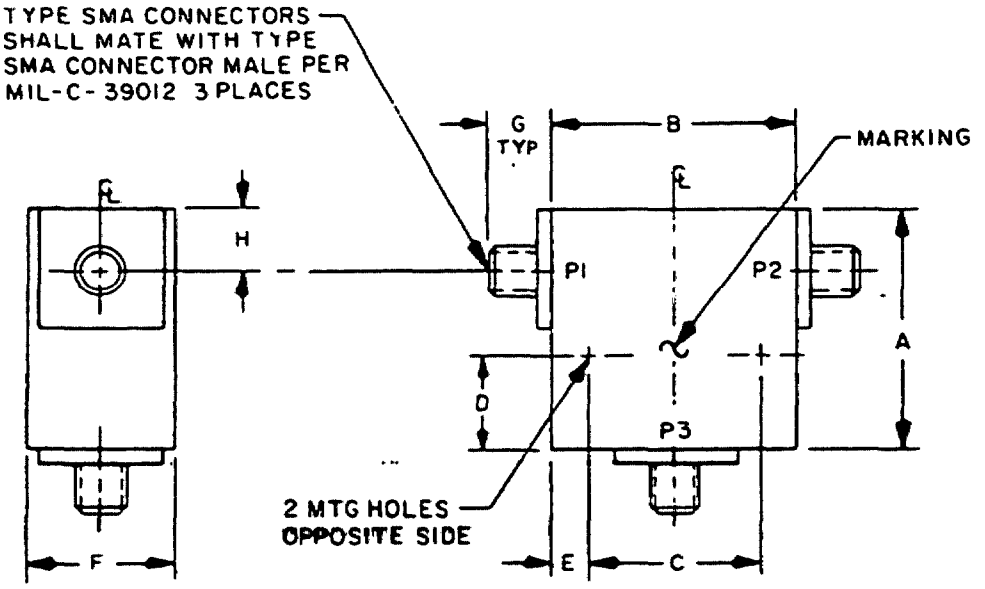
MILITARY SPECIFICATION SHEET

CIRCULATORS, RADIO FREQUENCY, COAXIAL (SERIES SMA), THREE PORTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the circulators described herein shall consist of this document and the latest issue of Specification MIL-C-28790.

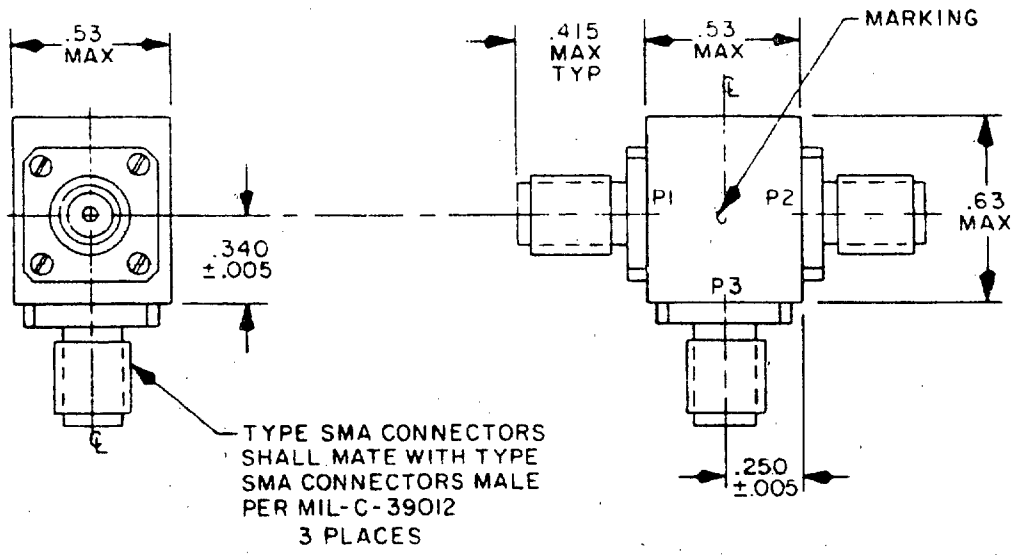


NOTES:
1. See table I for dimensions.
2. All undimensioned pictorial representations are for reference purposes only.
3. H dimension applies to parts P1 and P2 only.

FIGURE 1. Dimensions and configuration for part number M28790/1-001 thru -005.

PSC 5985

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INCHES	MM
.005	.13
.250	6.35
.340	6.64
.415	10.54
.53	13.5
.63	16.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Dimensions and configuration for part number M28790/1-006.

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TABLE I. Dimensional and thread requirements.

Dash no.	Dimensions 1/ 2/													MIG threads
	A	B	C		D		E		F	G		H		
	Max	Max	Min	Max	Min	Max	Min	Max	Max	Min	Max	Min	Max	
-001	.66 (16.8)	.53 (13.5)	.345 (8.76)	.355 (9.02)	.407 (10.34)	.417 (10.69)	.070 (1.78)	.080 (2.03)	.53 (13.5)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)	.086-56 UNC-2B x .13DP
-002	.66 (16.8)	.53 (13.5)	.345 (8.76)	.355 (9.02)	.407 (10.34)	.417 (10.69)	.070 (1.78)	.080 (2.03)	.53 (13.5)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)	.086-56 UNC-2B x .13DP
-003	1.055 (26.80)	.955 (24.26)	.645 (16.38)	.655 (16.64)	.395 (10.03)	.405 (10.29)	.145 (3.68)	.155 (3.94)	.65 (16.5)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)	.112-40 UNC-2B x .15DP
-004	1.055 (26.80)	.955 (24.26)	.645 (16.38)	.655 (16.64)	.395 (10.03)	.405 (10.29)	.145 (3.68)	.155 (3.94)	.65 (16.5)	.220 (5.59)	.415 (10.54)	.245 (6.22)	.255 (6.48)	.112-40 UNC-2B x .15DP
-005	1.23 (31.2)	1.23 (31.2)	.940 (23.88)	.960 (24.38)	.37 (9.4)	.43 (10.9)	.125 (3.18)	.135 (3.43)	.78 (19.8)	.35 (8.9)	.38 (9.7)	.23 (5.8)	.29 (7.4)	.0860-56 UNC-2B x .14DP

1/ Metric equivalents are in parentheses.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

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ENGINEERING DATA:

Frequency range: See table II.
Voltage rating (sea level): 335 V.
Nominal impedance: 50 ohms.
Operating temperature range: See table II.
Power rating: See table II.

Circulation: 1 → 2 → 3 → 1.
RF connectors: Shall be passivated stainless steel.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2 and table I.
Dielectric withstanding voltage: 1,000 volts rms, minimum.
Isolation: See table II.
Insertion loss: See table II.
VSWR: See table II.
Salt spray (for dash number -006 only):
MIL-STD-202, method 101, test condition B. This test shall be performed under first article inspection after the vibration test. Insertion loss and VSWR shall be measured after the test.
Marking: See figures 1 and 2.
Part number: M28790/1-(dash number from table II).

TABLE II. Electrical requirements.

Dash number	Frequency range	Isolation	Insertion loss	VSWR	Operating temperature range	Power rating
	GHz	dB(min)	dB(max)	Max		Max
-001	9.50-10.0	20	0.3	1.25:1	-54° to +95°C	10 watts
-002	12.5-13.5	20	0.3	1.25:1	↓	↓
-003	7.0-11.0	20	0.5	1.30:1		
-004	11.0-18.0	20	0.5	1.30:1		
-005	0.95-1.225	20	0.5	1.25:1	-54° to +85°C	50 watts
-006	9.0-9.2	20	0.3	1.25:1	-29° to +70°C	1 watt

NOTE: Circulators covered by this specification sheet are substituted for the following manufacturer's part numbers. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	Manufacturer's code	Manufacturer's part number	FSN
-005	81755	C9398-1	
-005	24022	C-0523T-11	
-006	49956	SPA-W-30310	5985-01-045-6963
-006	96341	M3E9200	5985-01-045-6963

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force - 11
DLA - ES

User activities:
Army -
Navy - OS, SH, MC
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0929)

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-C-28790/1A Circulators, Radio Frequency, Coaxial (Series SMA), Three Ports

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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